

Part A – NumPy

Question 1

Create a NumPy array containing the integers from 1 to 20.

Perform the following operations:

1. Display the array.
2. Find the shape of the array.
3. Find the maximum and minimum values.
4. Calculate the mean and standard deviation.
5. Reshape the array into a 4×5 matrix.
6. Display the second row of the matrix.
7. Display the third column of the matrix.

Question 2

Create two NumPy arrays:

A = [10, 20, 30, 40, 50]

B = [5, 10, 15, 20, 25]

Perform:

1. Addition of A and B.
2. Subtraction of B from A.
3. Element-wise multiplication.
4. Element-wise division.
5. Find the sum of all elements in A.

Part B – Pandas

Question 3

Create a DataFrame using the following student data:

Roll No Name Marks

101	Ravi	85
102	Priya	92
103	Kiran	78
104	Anu	88
105	Sita	95

Perform the following operations:

1. Display the DataFrame.
2. Display the first three records.
3. Find the average marks.
4. Find the student with the highest marks.
5. Find the student with the lowest marks.
6. Add a new column called "Grade" using:
 - A for Marks ≥ 90
 - B for Marks between 80 and 89
 - C for Marks < 80

Question 4

Create a DataFrame containing the following information:

Product	Price	Quantity
Laptop	50000	10
Mouse	500	50
Keyboard	1200	30
Monitor	10000	15

Perform:

1. Calculate the total value of each product using:

$$\text{Total Value} = \text{Price} \times \text{Quantity}$$

2. Add the Total Value column.
3. Find the product with the highest total value.
4. Calculate the overall inventory value.

Part C – Dataset Exploration

Question 5

Download the Iris dataset from Scikit-learn and perform the following:

1. Load the dataset into a Pandas DataFrame.
2. Display the first five records.
3. Find the number of rows and columns.
4. Display column names.
5. Check for missing values.
6. Generate summary statistics using `describe()`.
7. Find the number of samples belonging to each class.